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DOCUMENT DESCRIPTION
Schematic Checklist for the LAN8720I, 24-pin QFN Package

 	SMSC 80 Arkay Drive Hauppauge, New York 11788	
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	SC471242	A

Schematic Checklist for LAN8720I

Information Particular to the 24-pin QFN Package

LAN8720I QFN PHY Interface:

1. TXP (pin 21): This pin is the transmit twisted pair output positive connection from the internal PHY. It requires a 49.9Ω , 1.0% pull-up resistor to VDDA (created from +3.3V). This pin also connects to the transmit channel of the magnetics.
2. TXN (pin 20): This pin is the transmit twisted pair output negative connection from the internal PHY. It requires a 49.9Ω , 1.0% pull-up resistor to VDDA (created from +3.3V). This pin also connects to the transmit channel of the magnetics.
3. For transmit channel connection and termination details, refer to Figure 1.
4. RXP (pin 23): This pin is the receive twisted pair input positive connection to the internal PHY. It requires a 49.9Ω , 1.0% pull-up resistor to VDDA (created from +3.3V). This pin also connects to the receive channel of the magnetics.
5. RXN (pin 22): This pin is the receive twisted pair input negative connection to the internal PHY. It requires a 49.9Ω , 1.0% pull-up resistor to VDDA (created from +3.3V). This pin also connects to the receive channel of the magnetics.
6. For receive channel connection and termination details, refer to Figure 2.
7. For added EMC flexibility in a LAN8720I design, the designer should include four low valued capacitors on the TXP, TXN, RXP & RXN pins. Low valued capacitors (22 pF or less) can be added to each line and terminated to digital ground. These components can be added to the schematic and should be designated as Do Not Populate (DNP).

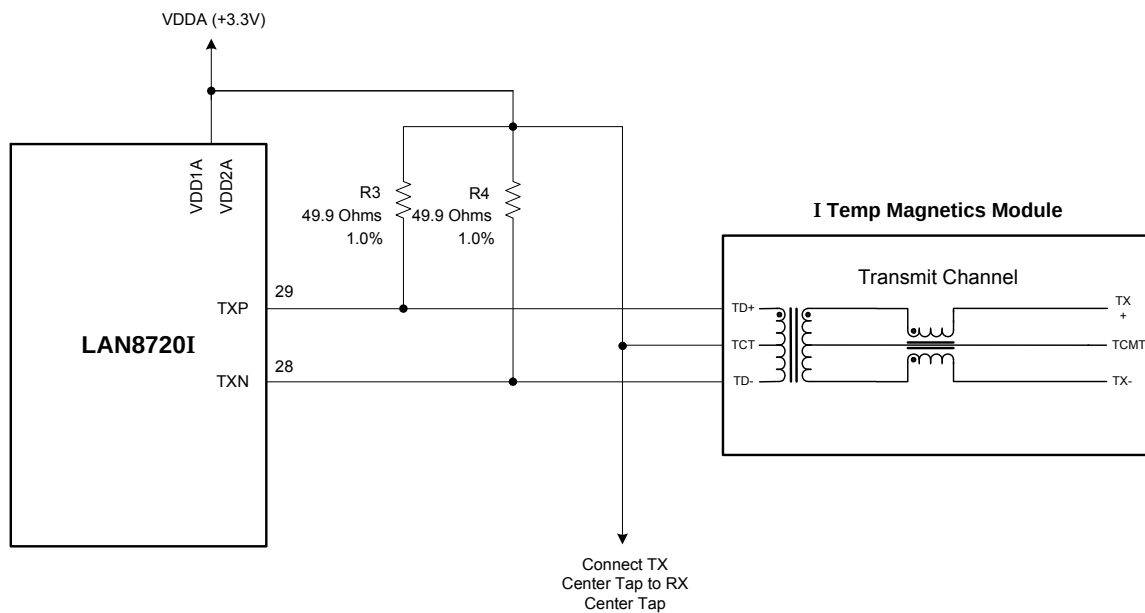


Figure 1 – Transmit Channel Connections and Terminations

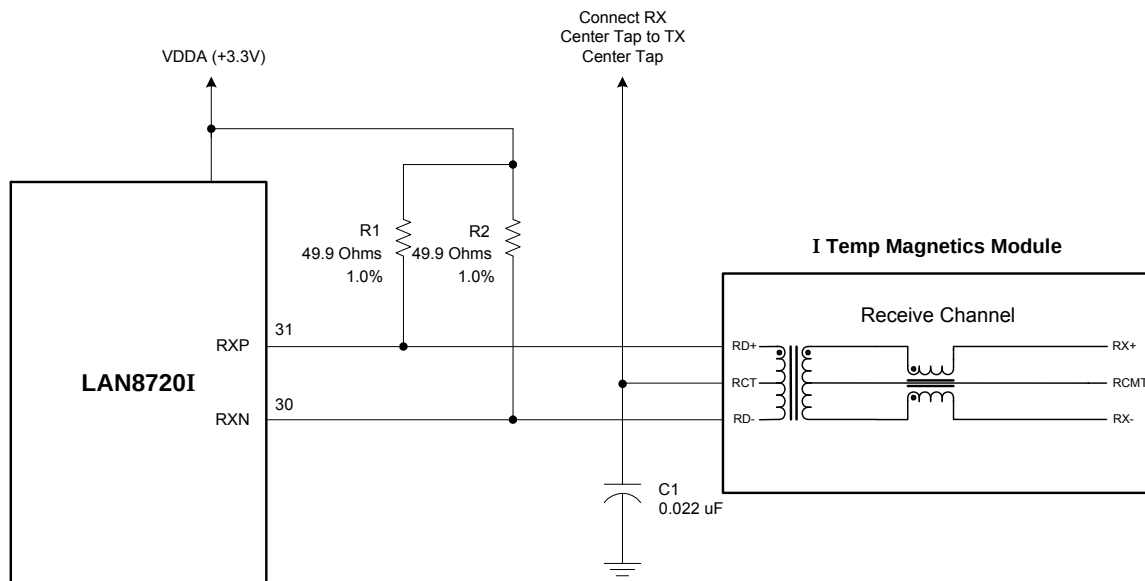


Figure 2 – Receive Channel Connections and Terminations

LAN8720I QFN Magnetics:

1. On the LAN8720I side, the transmit channel center tap connection must be connected to VDDA (created from +3.3V) directly. The transmit channel center tap of the magnetics also connects to the receive channel center tap of the magnetics.
2. On the LAN8720I side, the receive channel center tap connection is connected to the transmit channel center tap on the magnetics. In addition, a 0.022 μ F capacitor is required from the receive channel center tap of the magnetics to digital ground.
3. On the cable side (RJ45 side), the transmit channel center tap connection should be terminated with a 75 Ω resistor through a 1000 μ F, 2KV capacitor ($C_{magterm}$) to chassis ground.
4. On the cable side (RJ45 side), receive channel center tap connection should be terminated with a 75 Ω resistor through a 1000 μ F, 2KV capacitor ($C_{magterm}$) to chassis ground.
5. Only one 1000 μ F, 2KV capacitor ($C_{magterm}$) to chassis ground is required. It is shared by both TX & RX center taps.
6. Assuming the design of an end-point device (NIC), pin 1 of the RJ45 is TX+ and should trace through the magnetics to TXP (pin 21) of the LAN8720I QFN.
7. Assuming the design of an end-point device (NIC), pin 2 of the RJ45 is TX- and should trace through the magnetics to TXN (pin 20) of the LAN8720I QFN.
8. Assuming the design of an end-point device (NIC), pin 3 of the RJ45 is RX+ and should trace through the magnetics to RXP (pin 23) of the LAN8720I QFN.
9. Assuming the design of an end-point device (NIC), pin 6 of the RJ45 is RX- and should trace through the magnetics to RXN (pin 22) of the LAN8720I QFN.
10. When using the SMSC LAN8720I device in the HP Auto MDIX mode of operation, the use of an Auto MDIX style magnetics module is required. Please refer to the SMSC Application Note 8.13 "Suggested Magnetics" for proper magnetics.
11. In order to guarantee IEEE compliancy over the entire temperature range of operation, the magnetics used in conjunction with the LAN8720I must be rated for Industrial Temperature use.

RJ45 Connector:

1. Pins 4 & 5 of the RJ45 connect to one pair of unused wires in CAT-5 type cables. These should be terminated to chassis ground through a 1000 μF , 2KV capacitor (C_{rjterm}). There are two methods of accomplishing this:
 - a) Pins 4 & 5 can be connected together with two 49.9 Ω resistors. The common connection of these resistors should be connected through a third 49.9 Ω to the 1000 μF , 2KV capacitor (C_{rjterm}).
 - b) For a lower component count, the resistors can be combined. The two 49.9 Ω resistors in parallel look like a 25 Ω resistor. The 25 Ω resistor in series with the 49.9 Ω makes the entire circuit behave like a 75 Ω resistor. So, by shorting pins 4 & 5 together on the RJ45 and terminating them with a 75 Ω resistor in series with the 1000 μF , 2KV capacitor (C_{rjterm}) to chassis ground, an equivalent circuit is created.
2. Pins 7 & 8 of the RJ45 connect to one pair of unused wires in CAT-5 type cables. These should be terminated to chassis ground through a 1000 μF , 2KV capacitor (C_{rjterm}). There are two methods of accomplishing this:
 - a) Pins 7 & 8 can be connected together with two 49.9 Ω resistors. The common connection of these resistors should be connected through a third 49.9 Ω to the 1000 μF , 2KV capacitor (C_{rjterm}).
 - b) For a lower component count, the resistors can be combined. The two 49.9 Ω resistors in parallel look like a 25 Ω resistor. The 25 Ω resistor in series with the 49.9 Ω makes the entire circuit behave like a 75 Ω resistor. So, by shorting pins 4 & 5 together on the RJ45 and terminating them with a 75 Ω resistor in series with the 1000 μF , 2KV capacitor (C_{rjterm}) to chassis ground, an equivalent circuit is created.
3. The RJ45 shield should be attached directly to chassis ground.

Power Supply Connections:

1. The analog supply (VDD1A & VDD2A) pins on the LAN8720I QFN are 1 & 19. They require a connection to VDDA (created from +3.3V through a ferrite bead). Be sure to place bulk capacitance on each side of the ferrite bead.

Note: Pins 1 & 19 (VDD1A & VDD2A) must always be connected to a +3.3V power supply; even in the case of having the internal +1.2V regulator of the LAN8720I disabled. Other blocks within the LAN8720I require power from +3.3V.

2. Each VDDxA pin should have one .01 μ F (or smaller) capacitor to decouple the LAN8720I. The capacitor size should be SMD_0603 or smaller.
3. Pin 9 (VDDIO) is a variable supply voltage for the I/O pads. This pin must be connected to a voltage supply between +1.8V and +3.3V. The VDDIO power plane should have proper bulk capacitance.
4. The VDDIO pin should have one .01 μ F (or smaller) capacitor to decouple the LAN8720I. The capacitor size should be SMD_0603 or smaller.

Ground Connections:

1. The digital ground pins (GND), the analog ground pins (AVSS), and the GND_CORE pins on the LAN8720I QFN are all connected internally to the exposed die paddle ground. The EDP Ground pad on the underside of the LAN8720I must be connected directly to a solid, contiguous digital ground plane.
2. It is recommended that all ground connections be tied together to the same ground plane. It is not recommended to run separate ground planes for any SMSC LAN products.

VDDCR:

1. VDDCR (pin 6) is used to provide bypassing for the +1.2V core regulator. This pin requires a 470 pF bypass capacitor. This capacitor should be located as close as possible to the pin without using vias. In addition, pin 6 requires a bulk capacitor placed as close as possible to the pin. The bulk capacitor must have a value of at least 1.0 μF , and have an ESR (equivalent series resistance) of no more than 1.0 Ω . SMSC recommends a very low ESR ceramic capacitor for design stability. Other values, tolerances & characteristics are not recommended.

Caution: This +1.2V supply is for internal logic only. **Do Not** power other circuits or devices with this supply.

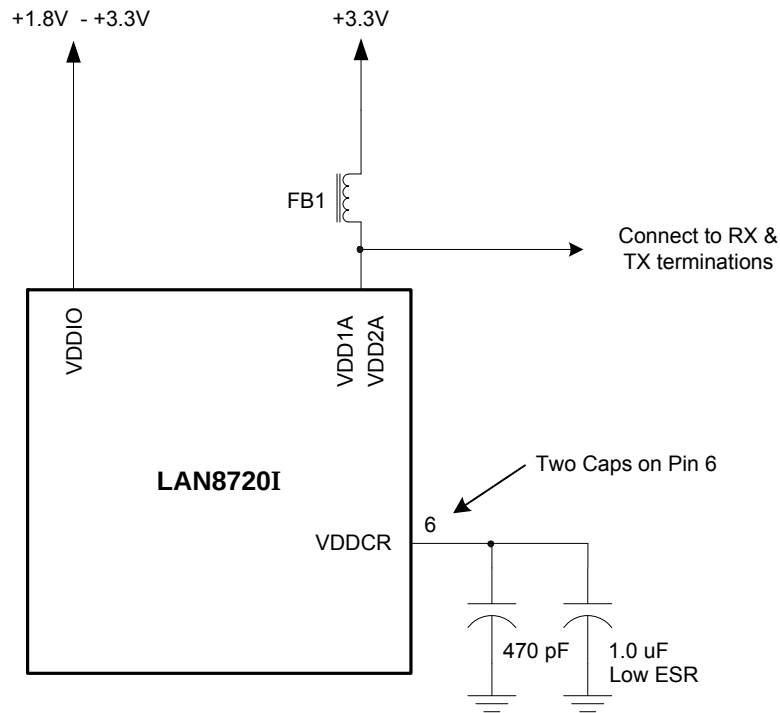


Figure 3 – LAN8720I Power Connections

Crystal Connections:

1. A 25.000 MHz crystal should be used to provide the clock source for the LAN8720I QFN. For exact specifications and tolerances refer to the latest revision of the LAN8720I data sheet.
2. XTAL1/CLKIN (pin 5) on the LAN8720I QFN is the clock circuit input. This pin requires a 15 – 33 pF capacitor to digital ground. One side of the crystal connects to this pin.
3. XTAL2 (pin 4) on the LAN8720I QFN is the clock circuit output. This pin requires a 15 – 33 pF capacitor to digital ground. One side of the crystal connects to this pin.
4. Since every system design is unique, the capacitor values are system dependant. The PCB design, selected crystal, layout, and the type of capacitors selected, all contribute to the characteristics of this circuit. Once the board is complete and operational, it is up to the system engineer to analyze this circuit in a lab environment. The system engineer should verify the frequency, stability, and voltage level of the circuit to guarantee that the circuit meets all design criteria as put forth in the data sheet.
5. An additional external 1.0M Ω resistor across the crystal is **not** required. The necessary resistance has been designed into the LAN8720I internally.
6. When using a 25.000 MHz crystal with the LAN8720I, the PHY generates the required 50.000 MHz for the RMII interface internally for its own use. A copy of the 50.000 MHz clock is provided as an output on pin 14 (nINT/REFCLK0) for use as the 50.000 MHz MAC REFCLK.
7. It is recommended that the designer use a series 33 Ω termination resistor on the REFCLK0 pin. The value can then be adjusted to compensate for any PCB trace impedance inconsistencies.
8. The REF_CLK Out Mode is not part of the RMII Specification. Timing in this mode is not compliant with the RMII specification. To ensure proper system operation, a timing analysis of the MAC and LAN8720I must be performed. Some MACs may require a small delay (500 pS – 1.0 nS) to the RXD[1..0] & CRS_DV signals. One method to achieve such a delay is to serpentine the signals from the Phy to the MAC.

9. In this application, nINTSEL must be a level zero during POR or nRST.
10. In order to guarantee IEEE compliancy over the entire temperature range of operation, the crystal used in conjunction with the LAN8720I must be rated for Industrial Temperature use.

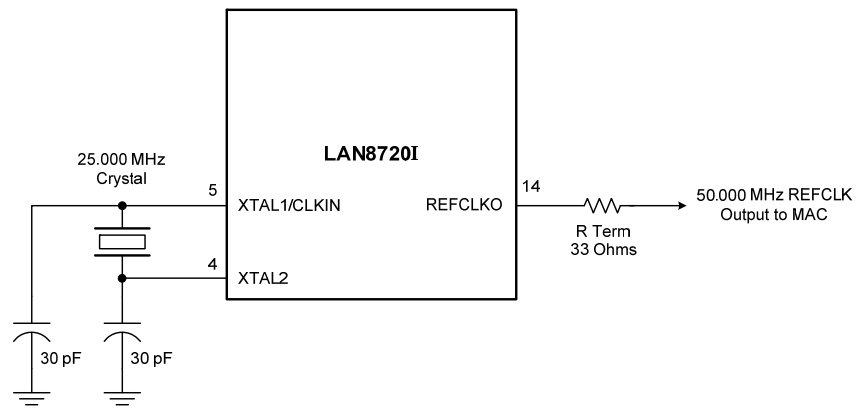


Figure 4 – LAN8720I Crystal Connections

Clock Oscillator Connections:

1. A 50.000 MHz clock oscillator may be used to provide the clock source for the LAN8720I. The clock oscillator must provide a 50.000 MHz clock for the PHY and RMII MAC in the design. For exact specifications and tolerances refer to the latest revision LAN8720I data sheet.
2. In order to provide two copies of the 50.000 MHz clock, it is recommended that the designer use two series 33 Ω resistors. The values can then be adjusted to compensate for any PCB trace inconsistencies.
3. XTAL1/CLKIN (pin 5) on the LAN8720I QFN is the clock circuit input. With low VDDIO voltages (+1.8V), CLKIN voltage may range from +1.8V to +3.3V.
4. XTAL2 (pin 4) on the LAN8720I QFN is the clock circuit output. When using a single ended clock source, this pin can be left floating as a No Connect (NC).
5. Since every system design is unique, the PCB design, oscillator selected, and layout all contribute to the characteristics of this circuit. Once the board is complete and operational, it is up to the system engineer to analyze this circuit in a lab environment. The system engineer should verify the frequency, stability, and voltage level of the circuit to guarantee that the circuit meets all design criteria as put forth in the data sheet.
6. In this application, nINTSEL must be a level one during POR or nRST.

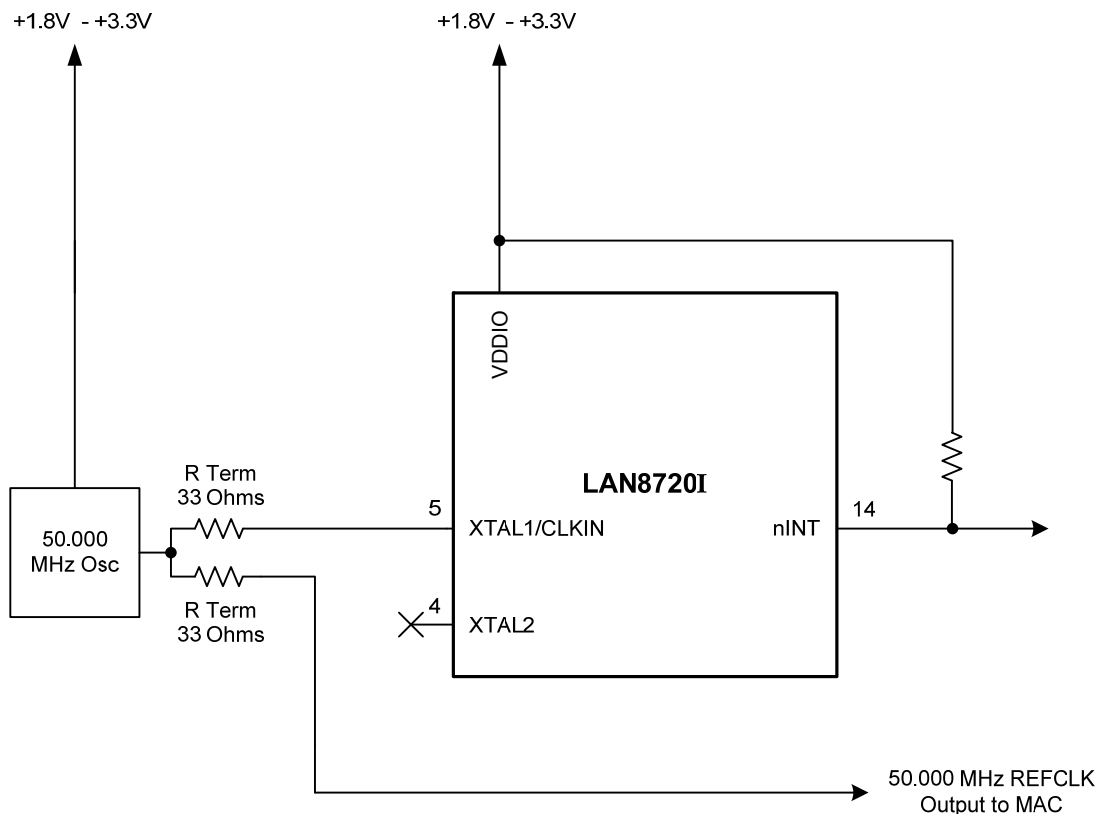


Figure 5 – LAN8720I Clock Oscillator Connections

MAC REFCLK Connections:

1. A 50.000 MHz REFCLK output from the MAC may be used to provide the clock source for the LAN8720I. For exact specifications and tolerances refer to the latest revision LAN8720I data sheet.
2. It is recommended that the designer use a series 33 Ω resistor at the MAC to connect to the Phy. The value can then be adjusted to compensate for any PCB trace inconsistencies.
3. XTAL1/CLKIN (pin 5) on the LAN8720I QFN is the clock circuit input. With low VDDIO voltages (+1.8V), CLKIN voltage may range from +1.8V to +3.3V.
4. XTAL2 (pin 4) on the LAN8720I QFN is the clock circuit output. When using a single ended clock source, this pin can be left floating as a No Connect (NC).
5. Since every system design is unique, the PCB design and layout all contribute to the characteristics of this circuit. Once the board is complete and operational, it is up to the system engineer to analyze this circuit in a lab environment. The system engineer should verify the frequency, stability, and voltage level of the circuit to guarantee that the circuit meets all design criteria as put forth in the data sheet.
6. In this application, nINTSEL must be a level one during POR or nRST.

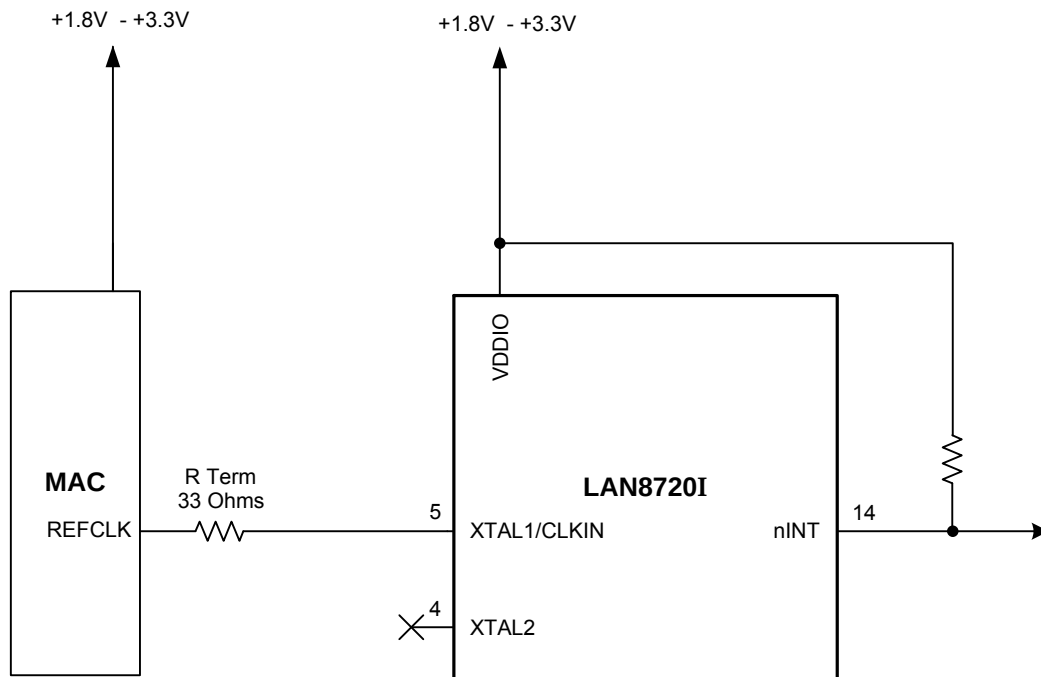


Figure 6 – LAN8720I MAC REFCLK Connections

RBIAS Resistor:

1. RBIAS (pin 24) on the LAN8720I QFN should connect to digital ground through a 12.1K Ω resistor with a tolerance of 1.0%. This pin is used to set-up critical bias currents for the embedded 10/100 Ethernet Physical device.

RMII Interface:

1. When utilizing an external RMII MAC interface, the following table indicates the proper connections for the 9 signals:

From:	Connects To:	
LAN8720I QFN	RMII MAC Device	Notes
RXD0 (pin 8)	RXD<0>	
RXD1 (pin 7)	RXD<1>	
RXD2	RXD<2>	Not Used in RMII Mode
RXD3	RXD<3>	Not Used in RMII Mode
RX_DV	RX_DV	Not Used in RMII Mode
RX_ER (pin 10)	RX_ER	This signal is optional in RMII Mode
RX_CLK	RX_CLK	Not Used in RMII Mode
TX_ER	TX_ER	Not Used in RMII Mode
TXD0 (pin 17)	TXD<0>	
TXD1 (pin 18)	TXD<1>	
TXD2	TXD<2>	Not Used in RMII Mode
TXD3	TXD<3>	Not Used in RMII Mode
TX_EN (pin 16)	TX_EN	
TX_CLK	TX_CLK	Not Used in RMII Mode
CRS_DV (pin 11)	CRS_DV	
CRS	CRS	Not Used in RMII Mode
COL	COL	Not Used in RMII Mode
MDIO (pin 12)	MDIO	
MDC (pin 13)	MDC	

2. Provisions should be made for series terminations for all outputs on the RMII Interface. Series resistors will enable the designer to closely match the output driver impedance of the LAN8720I and PCB trace impedance to minimize ringing on these signals. Exact resistor values are application dependant and must be analyzed in-system. A suggested starting point for the value of these series resistors is 10.0 Ω .

RMII Series Terminations:

	Series Terminations	
Signal	RMII Mode	Notes
RXD0	10 Ω	
RXD1	10 Ω	
RX_ER	10 Ω	
CRS_DV	10 Ω	

Required External Pull-ups:

1. Because the nINT (pin 14) output is open drain, an external pull-up resistor to VDDIO is required.
2. When using the RMII interface of the LAN8720I QFN with a MAC device on board, a pull-up resistor on the MDIO signal (pin 12) is required. A pull-up resistor of 1.5K Ω to VDDIO is required for this application.

Mode Pins:

1. The Mode pins of the LAN8720I (MODE[2:0]) control the default configuration of the 10/100 PHY. Speed, Duplex, Auto-Negotiation & power down functionality can be configured through these pins. The value of these three pins are latched upon power-up and reset. The latched values are reflected in Register 0 & Register 4 of the LAN8720I. Refer to the LAN8720I data sheet for complete details for the operation of these pins. These three pins have weak internal pull-ups and can be left as no-connects. To set any Mode bit low, an external 10K pull-down resistor should be used.

PHY Address Pins:

1. The PHY Address pin of the LAN8720I (PHYAD0) determines which of the 2 PHY addresses, of the 32 possible, the LAN8720I will respond to. The value of this pin is latched upon power-up and reset. The latched value is reflected in Register 18 of the LAN8720I. Refer to the LAN8720I data sheet for complete details on the operation of this pin. This pin has a weak internal pull-down and can be left as no-connect. To set the PHY Address bit high, an external 10K pull-up resistor to VDDIO should be used. Address bits PHYAD1, PHYAD2, PHYAD3 & PHYAD4 are tied low inside the LAN8720I.
2. A basic PHY Address of 01h is usually recommended.
3. The PHY Address pin is shared with an MII signal on the LAN8720I. The pinout is as follows:

PHY Address 0 is shared with RX_ER on pin 10.

PHY Address 1 is tied low.

PHY Address 2 is tied low.

PHY Address 3 is tied low.

PHY Address 4 is tied low.

LED Pins:

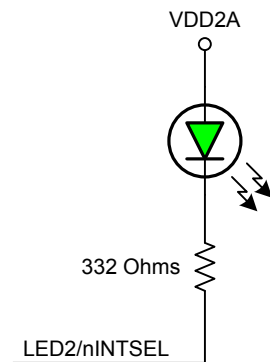
1. The LAN8720I provides two LED signals. These indicators will display speed, link and activity information about the current state of the PHY. The LED outputs have the ability to be either active high or active low. The polarity is determined by the level latched at nRST or POR. The LAN8720I senses each strap level value and changes the polarity of the LED signal accordingly. If the strap value is set as a level one, the LED polarity will be set to an active-low. If the strap value is set as a level zero, the LED polarity will be set to an active high. Refer to the LAN8720I data sheet for further details on how to strap each pin for correct operation and LED polarity outcomes.
2. The LED functionality signal pins are shared with the REGOFF & nINTSEL functionality of the LAN8720I. The pinouts are as follows:

LED1 is shared with REGOFF on pin 3.

LED2 is shared with nINTSEL on pin 2.

Interrupt Functionality:

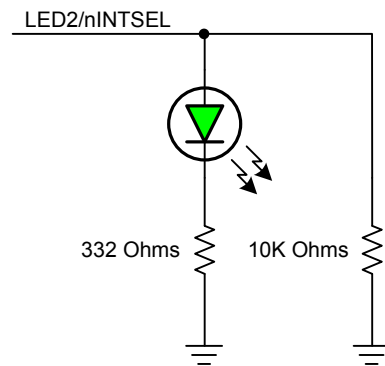
1. For added flexibility, the LAN8720I QFN provides a discrete interrupt line for embedded applications. This is advantageous as there is no interrupt facility across the standard MII Bus interface.
2. The nINT pin (pin 14) provides the interrupt signal from the LAN8720I. To enable the interrupt functionality on pin 14, the LED2/nINTSEL pin (pin 2) must be left as a no-connection. The LED2/nINTSEL pin has a weak internal pull-up and therefore can be left as a no-connect to select the interrupt functionality. The LED2/nINTSEL level is latched in on POR or nRST.
3. When the LED2/nINTSEL pin (pin 2) is used in conjunction with a LED, refer to Figure 6 below for details.



nINTSEL Bit = 1

**LED Output Signal from
LAN8720I is Active Low**

**Interrupt Functionality
Selected for Pin 14**



nINTSEL Bit = 0

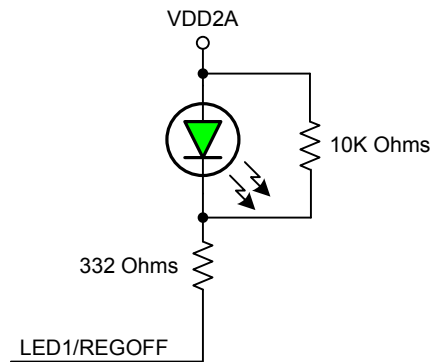
**LED Output Signal from
LAN8720I is Active High**

**REFCLKO Functionality
Selected for Pin 14**

Figure 7 – Interrupt Select / LED Polarity

Miscellaneous:

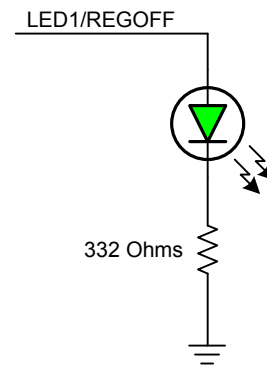
1. REGOFF (pin 3) enables/disables the internal +1.2V core regulator of the LAN8720I. This pin has a weak internal pull-down and can be left as a no-connect to enable the internal +1.2V regulator. To disable the +1.2V regulator, this pin should be pulled high with a 10.0K resistor to VDD2A. The REGOFF level is latched in on POR only.
2. When the LED1/REGOFF pin (pin 3) is used in conjunction with a LED, refer to Figure 7 below for details.



REGOFF Bit = 1

**LED Output Signal from
LAN8720I is Active Low**

**Internal +1.2V Core Regulator
is Disabled**



REGOFF Bit = 0

**LED Output Signal from
LAN8720I is Active High**

**Internal +1.2V Core Regulator
is Enabled (Default)**

Figure 8 – REGOFF / LED Polarity

Miscellaneous:

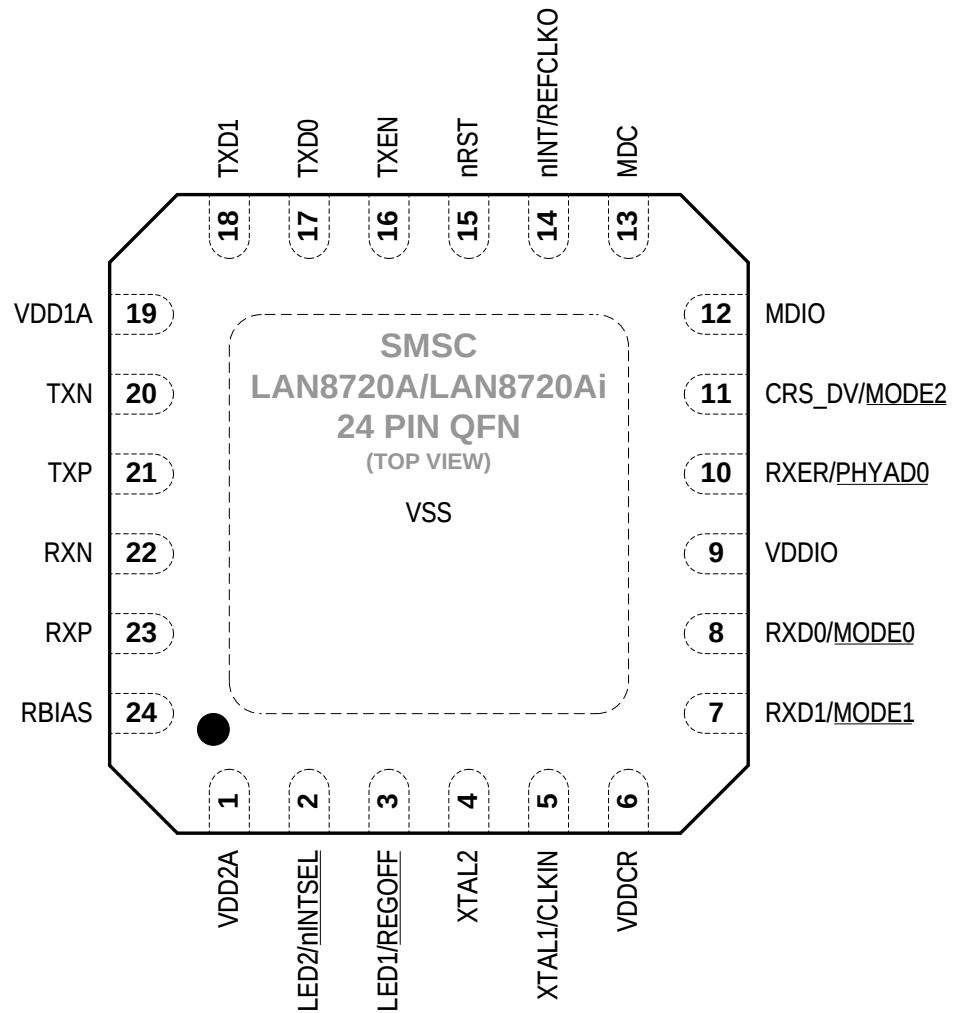
3. The nRST pin (pin 15) is an active-low reset input. This signal resets all logic and registers within the LAN8720I. This pin has a weak internal pull-up termination. A hardware reset (nRST assertion) is required following power-up. Please refer to the latest copy of the LAN8720I data sheet for reset timing requirements. SMSC does not recommend the use of an RC circuit for this required pin reset. A reset generator / voltage monitor is one option to provide a proper reset. Better yet, for increased design flexibility, a controllable reset (GPIO, dedicated reset output) should be considered. In this case, SMSC recommends a push-pull type output (not an open-drain type) for the monotonic reset to ensure a sharp rise time transition from low-to-high.
4. Due to possible lower I/O voltages used on the LAN8720I, lower strapping resistor values need to be used to ensure the strapped configuration is properly latched into the PHY device upon power-on reset. Refer to the latest revision of the LAN8720I QFN data sheet for details of proper resistor values when using lower I/O voltages on VDDIO.
5. Incorporate a large SMD resistor (SMD_1210) to connect the chassis ground to the digital ground. This will allow some flexibility during EMI testing for different grounding options in order to determine the best performing configuration:
 - o Leave the footprint blank for two separate ground planes
 - o Short the ground planes together at a single point with a zero ohm resistor
 - o AC couple the grounds together with a high voltage capacitor
 - o Connect the two planes together with a ferrite bead
6. Be sure to incorporate enough bulk capacitors (4.7 - 22 μ F caps) for each power plane.
7. In order to guarantee IEEE compliancy over the entire temperature range of operation, all components used in the customer's application must be rated for Industrial Temperature use. Processors, crystals, oscillators, magnetics and all integrated circuits must be rated properly to avoid operational inconsistencies.

LAN8720I QFN QuickCheck Pinout Table:

Use the following table to check the LAN8720I QFN shape in your schematic:

LAN8720I QFN			
Pin No.	Pin Name	Pin No.	Pin Name
1	VDD2A	13	MDC
2	LED2/nINTSEL	14	nINT/REFCLKO
3	LED1/REGOFF	15	nRST
4	XTAL2	16	TXEN
5	XTAL1/CLKIN	17	TXD0
6	VDDCR	18	TXD1
7	RXD1/MODE1	19	VDD1A
8	RXD0/MODE0	20	TXN
9	VDDIO	21	TXP
10	RXER/PHYAD0	22	RXN
11	CRS_DV/MODE2	23	RXP
12	MDIO	24	RBIAS
25		EDP Ground Connection Exposed Die Paddle Ground Pad on Bottom of Package	

LAN8720I QFN Package Drawing:



NOTE: Exposed pad (VSS) on bottom of package must be connected to ground

Reference Material:

1. SMSC LAN8720I Data Sheet; check web site for latest revision.
2. SMSC LAN8720 CEB Schematic, Assembly No. 6584; check web site for latest revision.
3. SMSC LAN8720 CEB PCB, Assembly No. 6584; order PCB from web site.
4. SMSC LAN8720 CEB PCB Bill of Materials, Assembly No. 6584; check web site for latest revision.
5. CEB stands for Customer Evaluation Board.
6. SMSC LAN8720I Reference Design, check web site for latest revision.
7. SMSC Reference Designs are schematics only; there are no associated PCBs.
8. For Qualified / Suggested Magnetics, use these two links to the SMSC LANCheck website:

https://www2.smisc.com/mkt/web_lancheck.nsf/MagList?OpenForm

https://www2.smisc.com/mkt/web_lancheck.nsf/MagCheck?OpenForm